Phisher - A Multimodal Approach for Phishing Detection

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Abstract

Phishing is an illegal method to trick people into revealing confidential information such as login details, credit card numbers, and social security numbers. The majority of these phishing activities are carried out by duplicating the appearance of authentic websites or emails and exploiting people's trust, rather than technical vulnerabilities. Although numerous awareness campaigns and technical countermeasures are aimed at alerting individuals about phishing, phishing remains one of the most effective methods of cyber assault due to its malleability and continually evolving complexity. Many single-modal models are effective to a certain degree, but cannot identify advanced phishing techniques that incorporate dynamic web content, obfuscated scripts, and sophisticated visual mimicry. We introduced a novel multimodal approach called Phisher. Our multimodal models utilize BERT Multimodal Large Language Model (MLLM) for combined lexical analysis, ResNet50 for image processing, and semantic characteristics for URL extraction to enhance phishing classification. URL features are highly efficient at detecting unusual domains and token anomalies; HTML features can detect hidden scripts or iframe attacks; and visual features aid in identifying brand impersonation or misleading UI features. By combining these signals, we can obtain a more accurate model, which aids in detecting phishing sites. To create our multimodal model, we utilized the TR-OP real-life dataset, which contains 10,000 phishing and legitimate websites of HTML content, URL, and website snapshots, and the results show significant improvement in accuracy and precision compared to other models.

1 Introduction

Phishing is perhaps the most widespread and destructive type of cybercrime these days. It describes a kind of social engineering attack wherein attackers disguise themselves as a trusted entity—institutions like financial institutions, government agencies, or online services—to trick individuals into divulging sensitive personal details. Most of these phishing attempts look like legitimate emails, mock websites, or SMS messages but are designed carefully to take advantage of human trust instead of a technical vulnerability. Attackers usually entice victims into clicking on dangerous links, entering login credentials, or installing malware (Wikipedia, 2024).

According to the FBI's Internet Crime Report, phishing was the most reported cybercrime in 2023, accounting for over 700,000 complaints (Federal Bureau of Investigation, 2024) and resulting in estimated financial losses exceeding $2.9 billion globally. The Anti-Phishing Working Group (APWG) also observed a record high in phishing attacks during Q4 2023, with over 1.4 million unique phishing websites detected in just three months. These statistics demonstrate that phishing is not only widespread but also growing in both scale and sophistication.

In order to counter this threat, many types of detection methods have been proposed over the years. Most conventional phishing detection methods rely on rule-based systems, blacklisting, or handcrafted lexical features. These perform fine for mature attacks but fail to detect new or obfuscated phishing attacks. Additionally, attackers now more frequently use sophisticated evasion methods like dynamic content generation, script obfuscation, and visual spoofing for which conventional methods become ineffective.

With the emergence of machine learning (ML) and artificial intelligence (AI) in recent years comes new promise for phishing detection too. ML-based models can be learned against large datasets of URLs, emails, or web pages. Static rule-based methods have been surpassed in terms of accuracy by methods like decision trees, random forests, support vector machines (SVMs), and newer deep learning methods. These methods largely base themselves on features solely from a single modality—for instance, examining textual content or the structure of URLs in isolation. Although single-modal models have their place, they possess significant drawbacks. Detectors like URL-based can identify domain names that appear suspicious, but cannot possibly take into consideration contextual or semantic information. Text-based models can be fooled with carefully constructed text or scripted embedding. Even vision-based methods can be fooled with subtle image manipulation. As sophistication in phishing increases, relying on a single input modality becomes increasingly inadequate for comprehending the diverse facets of phishing websites.

In order to address these challenges, focus has now been directed towards multimodal phishing detection where various types of information—such as structure of the URL, HTML content, and visual aspects—are combined together for more reliable and solid prediction. These modalities offer distinct levels of insight:URL features identify anomalous tokens, domain anomalies, or suspi-

cious patterns. HTML features help identify hidden elements, malicious JavaScript, or fake forms. Visual features (e.g., screenshots) detect branding mismatches, layout forgery, and UI copying. In this paper, a multimodal detector for phishing is proposed which combines all three interaction modes for more accurate classification. We design our method using the TR-OP (Threat Report - OpenPhish) dataset with 10,000 labeled instances of the web along with corresponding screenshots, HTML source code, and URLs. For basic-level text analysis, TF-IDF vectors along with BERT embedding vectors are used for identifying shallow and deep patterns semantically. Visual features are detected with the help of convolutional neural networks (CNNs) like ResNet50, whereas structure in the URLs is converted into a set of vector representations using handcrafted features (static) along with statistical features (learned). Combining these three views with each other, our detector is more accurate with enhanced resistance towards advanced phishing attacks.

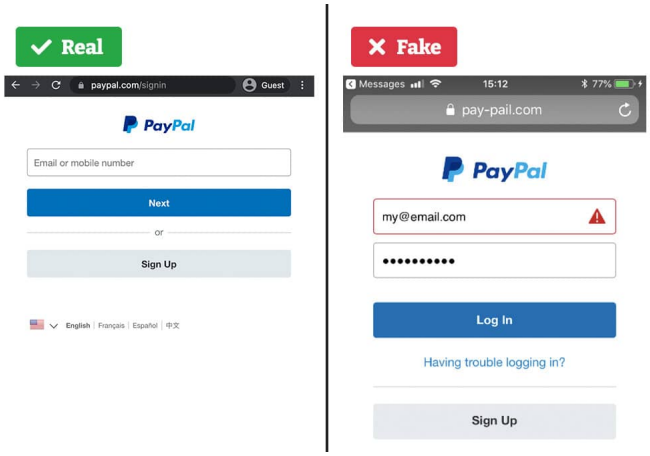


Figure 1: Comparison of a genuine PayPal login page and a phishing attempt. The left side (labeled "Real") demonstrates the real PayPal log-in screen, using the Correct domain paypal.com/signin. The right side (labeled "Fake") copies PayPal's style but adopts a deceptive domain paypail.com, which has the primary intention of duping users into validating their qualifications. The case illustrates the importance of checking the website's URL before supplying sensitive information.

1.1 Related Works

There are many approaches to detecting phishing websites. Some of these traditional methods include conventional processes, such as the blacklist/whitelist approach, the heuristic approach, and keyword detection approaches. With the recent advent of AI/ML models, numerous new and advanced methods for detecting phishing websites have emerged.

1.1.1 Blacklist or Whitelist approach

Blacklists contain known phishing URLs or domains, enabling detection and prevention of these URLs and domains from causing harm. Similar to blacklists, whitelists also contain URLs and domains, but they are known to be safe and reliable. The blacklist/whitelist approach is a reasonably secure method, but some loose ends can be exploited. For example, this approach can only be practical when there are already-identified threats. These methods are reactive and cannot mitigate modern-day phishing primarily because new phishing URLs or domains may not be listed. Another reason why this approach falls short of modern-day phishing detection is that blacklists and whitelists require constant maintenance and are susceptible to rapid domain churn by attackers[1].

1.1.2 Heuristic-Based Approaches

Heuristic-based Detection is the detection of phishing using handcrafted rules from domain knowledge. Some common heuristics include suspicious URL patterns, the use of HTTPS or SSL certificates to demonstrate authenticity, and excessive form elements or hidden input fields within HTML structures[2]. These rules might be easy to understand, but they usually suffer from rigidity. This means that people creating phishing material can easily bypass these rules by altering webpage elements, resulting in high false negatives if the heuristics are not regularly monitored.

1.1.3 Keyword-Based Detection

Keyword-Based Detection contains systems that scan webpages for specific terms related to phishing, such as "login", "verify account", and other key associated terms[3]. These keywords usually appear in URLs, HTML content, or JavaScript. Although this method is effective, it is prone to false positives when safe pages contain similar terms. This method is also prone to adversarial evasion, a technique in which attackers conceal keywords by utilizing Unicode tricks or typosquatting.

1.1.4 Reference-Based Approaches

Reference-based methods typically create a Knowledge Base comprising various brands, including actual domains, logos, and aliases for popular brands. After scanning a new page, the content will be compared to the original base to determine whether it is a fake copy or not[4]. Even with high accuracy for known brands, scalability issues and coverage gaps arise when dealing with local or emerging brands not present in the brand database.

1.1.5 Search Engine-based Techniques

Search Engine-based techniques are methods that generate queries from webpage content, like domain names and HTML text, and analyze the results from the search engine to determine whether the page appears to be a legitimate result. If it is not legitimate, then the page will be flagged as malicious. These methods are still too sensitive to search engine indexing delays. They're also sensitive to changes in algorithms, and they can produce false positives for less-known safe websites[5].

1.1.6 LLM/MLLM-Based Approaches

Large Language models and Multimodal Large language models have recently been used to process textual and visual input for phishing detection. Some examples of these inputs include screenshots from the website and HTML data. These models are significantly stronger than their predecessors, and they can achieve high accuracies in detecting spoofed brands and content manipulation[6]. One flaw with these models is that they are still vulnerable to adversarial attacks. These models also misinterpret pages targeting underrepresented/local brands due to limitations in training data.

1.1.7 Agent-Based Approaches

To help prevent the problems from the LLM/MLLM-based approaches, Agent-Based methods combine LLMs with toolkits and knowledge bases. In doing so, this enables decision-making in various modalities and also provides better adaptiveness against unseen phishing strategies[7]. In general, it increases accuracy and resilience to evasion techniques.

1.2 Multimodal Training Models

The proposed multimodal phishing detector significantly advances existing phishing site classification using a multiplicity of sources of information—i.e., HTML textual information, visual page screenshots, and lexical characteristics of URLs. In contrast with prior single-modality systems devoted exclusively to a single type of signal, a multimodal system aggregates complementary indications and therefore improves the accuracy and strength of the phishing detector for complex real-world applications[8].

1.2.1 Improved Detection Rates

Single-modal techniques—although operating for a single instance—will generally not generalize across the extensive set of phishing techniques. For example, a classifier trained using sole lexical features of URLs can detect anomalous URLs like http://login-paypal-verification.com correctly. Still, it will not detect a visually deceptive page sent off a compromised legitimate domain. A text-only technique will not be able to detect phishing pages that obscure text within the image or employ encoded scripts. Through an integration of textual, visual, and URL features at a multimodal level, the multimodal model can cross-reference between sources for any single signal. For instance, if the HTML has suspicious text such as "validate account" and the visual layout is very similar to a PayPal login layout, even a seemingly innocent-looking URL like https://secure.login-center.com is flagged correctly. This integration minimizes false negatives (phishing going undetected) and false positives (legitimate content being flagged as phishing). Empirical evidence confirms this strength. A multimodal phishing detector called PhishAgent[9] which enhanced overall F1-scores 7–10 % relative to single-modal detectors. Our experiments indicate multimodality achieves 98.13% accuracy, which is better than current baselines like Phishpedia[10] and GEPAgent[11] at most 3.5%.

1.2.2 Resistance to Sophisticated Attacks

Phishing websites attempt to keep a low profile by focusing their deceptions on a single modality. A phishing page, for instance, can sport a neat and innocent-looking URL but pack malicious scripts into the HTML, or vice versa, sport well-crafted text but a suspicious-looking structure for the URL. These deceptions aim at precluding systems from checking a single layer of data. A multimodal defense is automatically more immune to such evasion methods. A phisher may disguise anomalies within the URL but deploys false branding or dubious text content, yet the textual or visual modality can still activate an alarm. For example, even if an attacker deploys https://amazon-check-secure.com, the use of manipulated Amazon logos or forms characteristic of phishing can be identified with the ResNet-based visual classifier. Its multiple-layer redundancy ensures that even an attempt at evasion through a single modality won't impact overall system performance. This multi-layer redundancy guarantees that an attempt at evasion over a single modality will not impact the system's overall functionality.

1.2.3 Improved Semantic Comprehension

Conventional keyword-centric or bag-of-words methods for text classification usually have difficulties dealing with semantic ambiguity and context sensitivity. Incorporation of Bidirectional Encoder Representations from Transformers (BERT) allows for deep semantic understanding of webpage text. In contradistinction with shallow models, BERT is capable of grasping deep sentence structure, contextual use, and subtle deception markers. In other words, a phishing message like "Please validate your account to prevent suspension" can be semantically equivalent to "We must validate your login for security purposes," even if the language is slightly different. BERT's transformer-based structure enables the model to understand this equivalence and mark both examples, whereas older models might overlook one. It's a basic contextual knowledge for recognizing phishing material, which is abusive psychologically, or legally seasoned language manipulation for a sense of panic or fear.

1.2.4 Visual Pattern Recognition

One of the most efficient methods of phishing is visual mimicry, which attackers often use in creating pages that look like they belong to reputable institutions. ResNet50 is a deep convolutional neural network that can extract fine-grained spatial hierarchies from images. By checking screenshots of websites, the visual modality can identify things such as:

- Logo cloning (e.g., PayPal, Apple, Microsoft),
- Form placements mimicking login screens,
- Font inconsistencies or improper alignment,

- Low-resolution or manipulated images.

For instance, a phishing page may use a blurred version of the Bank of America logo or a layout with unusually large "Submit" buttons to trick users. Even if the textual or URL indicators are mild, the visual features alone could expose the attack. Our visual embedding approach improves precision in such cases, with ResNet-based models achieving up to 98.5% precision in recognizing high-risk mimicry patterns[12].

1.2.5 Thorough URL Analysis

Phishing URLs also often carry anomalies like atypical subdomain patterns, overuse of numbers, or the inclusion of IP addresses rather than domain names. Although certain URLs may look superficially innocuous at a glance, more in-depth lexical analysis can identify manipulation tactics. Specifically, features such as URL length, number of digits, hyphens, subdomains, and the presence of an IP address are key indicators of malicious intent[13].

URL features within our system are:

- **Length** (L_U): Longer URLs often aim to conceal suspicious tokens.
- **Digits** (D_U): Numeric values can simulate transaction IDs or user credentials.
- **Periods/Slashes** (P_U, S_U): Used for deep nesting or subdomain spoofing.
- **Hyphens** (H_U): Introduce deceptive similarity (e.g., `bank-login.com`).
- **IP Presence** (IP_U): Phishing often uses raw IPs instead of domain names.
- **HTTPS Indicator** (HTTPS_U): Some attackers now use HTTPS to appear legitimate.

Even sophisticated phishing attempts usually reveal at least a single anomaly within the URL. Encoding these lexical features enables the model to recognize malicious activity beyond what appears on the surface.

1.2.6 Generalization and Adaptability

Phishing tactics continuously evolve with time. Attackers modify language, image content, domain name, and layout pattern in an attempt to bypass static rule-based detection. Most single-modality systems must be retrained regularly or require feature engineering for change accommodation.

Multimodal systems, on the other hand, provide better generalization. Regardless of whether phishers refine a single modality (e.g., employing grammatically correct text), overall detection effectiveness is preserved because the other modalities provide strong signals themselves. This design provides for longevity of adaptability as well as a lessened reliance on a single threat indicator.

For instance, a 2023 paper about building privacy-preserving and secure AI foundation models demonstrated that multimodal models maintain more than 95% accuracy even if a single modality is corrupted or adversarially occluded, which is a far higher rate than 70–80% for individual modality systems[14].

2 Methods

Current phishing detection systems, whether rule-based, single-modal ML, or keyword-focused ones, have real shortcomings in adaptability, coverage, and resilience. Standard blacklists could not detect new or zero-day phishing domains. Heuristics are fragile and can be easily evaded using basic evasion methods. Vision-only or text-only ML systems break down when attackers include text in images, employ visually obfuscating but rightful-looking formats, or cunningly mask malicious intention in the URL. Even modern LLM-founded models, though robust, remain susceptible to adversarial examples and tend to classify attacks on less-represented brands incorrectly.

To counter such limitations, we introduce a fully multimodal system—Phisher, is a multimodal detector for phishing, employing textual feature extraction using BERT embeddings, visual feature extraction with ResNet50, and URL lexical feature engineering for robust binary classification of sites as harmless or phishing. With the integration of these heterogeneous data sources under a single unifying neural network classifier, the system significantly increases detection accuracy and is in a position to detect subtle phishing attacks that remain undetected under single-modality approaches. Each modality gives a strong representation of the respective input. Then, these embeddings are concatenated into a single feature vector for the classification model. Through experimentation with neural networks, Random Forest, and XGBoost for prediction, we found that neural networks achieve the highest accuracy, precision, recall, and F1-score across datasets. By combining multiple modalities, our model shows more robustness and generalizability than unimodal models. The multimodal architecture was extremely effective in overcoming adversarial evasion techniques, and our model is able to adapt efficiently to new phishing strategies.

2.1 Textual Embedding (BERT)

To capture the semantic and contextual information embedded in the HTML content of a webpage, we utilize the Bidirectional Encoder Representations from Transformers (BERT) model. BERT is well-suited for this task due to its ability to understand the deep bidirectional context of language and code structures within HTML.

Given the HTML content H, the BERT embedding E_{HTML} is defined as:

$$E_{\text{HTML}} = \text{BERT}(H), \quad E_{\text{HTML}} \in R^{768} \tag{1}$$

BERT computes embeddings through a series of Transformer encoder layers. These encoders capture both syntactic and semantic dependencies in the tokenized HTML. We extract the final pooled output from the transformer as the representative embedding vector:

$$E_{\text{HTML}} = \text{PoolerOutput}(\text{Transformer}_{\text{encoder}}(\text{Tokenizer}(H))) \tag{2}$$

This textual embedding captures both surface-level token patterns (e.g., suspicious words like "login", "account", "verify") and deeper semantic signals (e.g., obfuscated malicious scripts or deceptive metadata).

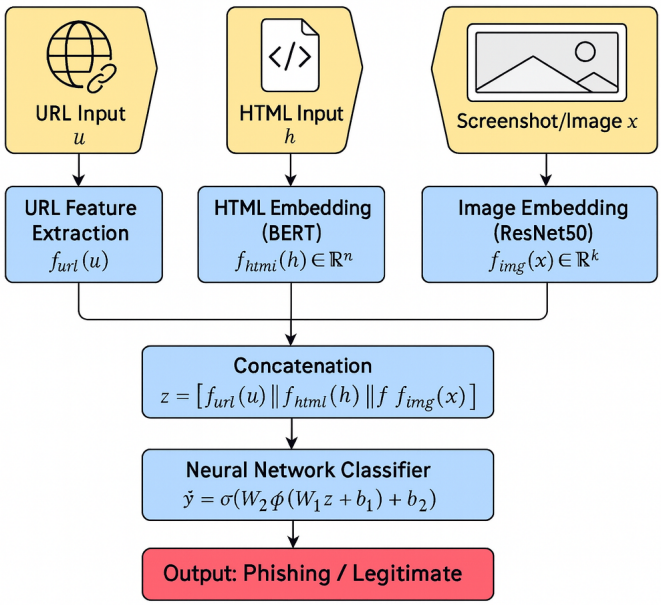


Figure 2: Architecture of the multimodal phishing detection model combining textual, visual, and URL features.

2.2 Visual Embedding (ResNet50)

Visual information from webpage screenshots plays a crucial role in identifying phishing attempts that mimic the appearance of legitimate websites. Phishing pages often replicate logos, buttons, and layouts of trusted organizations, which can be effectively captured using convolutional neural networks (CNNs).

For the screenshot image I, ResNet50 generates the embedding E_{IMG}:

$$E_{\text{IMG}} = \text{ResNet50}(I), \quad E_{\text{IMG}} \in R^{2048} \tag{3}$$

Specifically, ResNet50 consists of deep residual blocks that enable learning of rich hierarchical features. The visual embedding is computed by applying a global average pooling operation to the final CNN feature maps:

$$E_{\text{IMG}} = \text{GlobalAvgPooling}(\text{CNNLayers}(I)) \tag{4}$$

This vectorized representation encodes global visual patterns—such as font, color schemes, UI alignment, and brand mimicry—that are often exploited in phishing.

2.3 URL Lexical Feature Extraction

URLs remain one of the most indicative elements of phishing attacks. Malicious URLs often include specific lexical patterns such as excessive use of numeric characters, unusually long domains, or misleading keywords.

The URL U lexical feature vector E_{URL} is:

$$E_{\text{URL}} = [L_U, D_U, P_U, S_U, H_U, IP_U, HTTPS_U] \tag{5}$$

$$E_{\text{URL}} \in R^7 \tag{6}$$

The features are defined as follows:

- L_U: URL length — Longer URLs are often used to obfuscate malicious intent.
- D_U: Number of digits — Excessive digits may indicate autogenerated or fake subdomains.
- P_U: Number of periods (dots) — Used to create misleading subdomains or deep URL nesting.
- S_U: Number of slashes — Indicates URL depth or directory structure manipulation.
- H_U: Number of hyphens — Often used to mimic legitimate domains (e.g., pay-pal.com).
- IP_U: Binary indicator (IP address presence: 0 or 1) — IP-based URLs are common in phishing sites.
- $HTTPS_U$: Binary indicator (HTTPS usage: 0 or 1) — While HTTPS is generally secure, its misuse in phishing sites is growing.

These lexical indicators are lightweight yet powerful features in identifying suspicious URLs.

2.4 Multimodal Feature Fusion

Each modality—textual, visual, and lexical—offers a unique perspective on the phishing detection problem. While individually useful, these modalities can complement one another when fused, providing a holistic view of the webpage.

The concatenated embedding E_{Concat} is:

$$E_{\text{Concat}} = [E_{\text{HTML}}; E_{\text{IMG}}; E_{\text{URL}}], \quad E_{\text{Concat}} \in R^{2823} \tag{7}$$

This fused feature vector combines: - Semantic representations from HTML content via BERT, - Visual patterns from webpage screenshots via ResNet50, and - Lexical structures from URLs via handcrafted features.

The integration of all three modalities strengthens the model's ability to generalize across various phishing techniques, particularly those that evade detection in single-modality systems.

2.5 Neural Network Classifier

The final classification of a webpage as phishing or legitimate is performed using a fully connected neural network. This model maps the high-dimensional fused feature vector into a probability score between 0 and 1 using a non-linear transformation.

Classification is performed using a fully-connected neural network:

$$\hat{y} = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot E_{\text{Concat}} + b_1) + b_2) \tag{8}$$

where:

- W_1, W_2: Weight matrices responsible for learning feature transformations.
- b_1, b_2: Bias vectors that help the network shift activation boundaries.

- $\text{ReLU}(x) = \max(0, x)$: Rectified Linear Unit activation for non-linearity.
- $\sigma(x) = \frac{1}{1+e^{-x}}$: Sigmoid activation function to squash outputs between 0 and 1.

This structure enables the classifier to learn complex decision boundaries and leverage the complementary strengths of each modality to accurately predict phishing instances.

3 Dataset and Results

The TR-OP dataset is a handcrafted test dataset designed primarily for multimodal phishing website detection. Its purpose is to reflect realistic phishing and legitimate website scenarios by combining visual, textual, and structural information. This is perfect for multimodal machine learning modeling for training and testing purposes. It's also suited for multimodal models that combine visual embeddings (e.g., from ResNet50 or ViT), textual embeddings (e.g., from BERT or TF-IDF), and URL feature vectors (custom or heuristic-based). The TR-OP dataset comprises 5000 labeled phishing websites and 5000 labeled legitimate websites. Screenshot captures of legitimate pages compared with phishing pages, typically generated with headless browsers, represent layout, brand abuse, and visual indicators. Extracted HTML source code or rendered text content from the webpage allows access to suspicious keywords, hidden forms, and script behaviors. The raw features of URLs, like length, occurrence of special characters, subdomain depth, and lexical patterns, are usually connected with phishing URLs.

We tested four machine learning models namely Random Forest, XGB Classifier, Naive Bayes, and Neural Networks, and measured them using accuracy, precision, recall, F1 score, and inference time as parameters. The highest accuracy was achieved by Neural Networks at 0.9813, then by Random Forest at 0.9768 and by Naive Bayes at 0.9707. For F1 score, Neural Networks and Random Forest were equal at 0.9800 and Naive Bayes lower at 0.9700.

XGB Classifier was the fastest model, only consuming 0.0003 seconds in inference, however, but was the model with the poorest accuracy (0.9576) and F1 score (0.9600), indicating a trade-off of predictive performance and speed. In general, Neural Networks gave the best predictive output, and Naive Bayes and Random Forest gave reasonable and balanced performance across all attributes. XGB Classifier, although with lower accuracies, could be the best model if we have time-sensitive applications and need the fastest model to be utilized.

Key Insights

- **Highest Accuracy:** Neural Networks (0.9813) slightly outperform Naive Bayes (0.9707).
- **Best F1 Score:** Neural Networks and Random Forest both achieve 0.98, with Naive Bayes close behind (0.97).

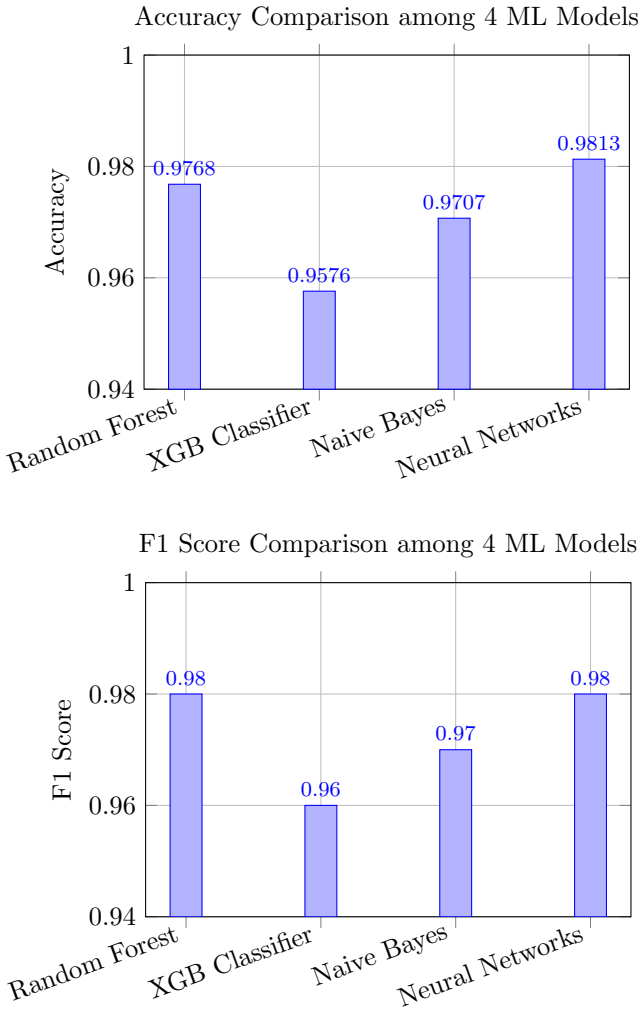


Figure 3: Performance comparison of our Model using different ML models based on Accuracy and F1 Score.

- **Fastest Model:** XGB Classifier (0.0003 s) significantly outpaces others in inference time, though it records the lowest accuracy and F1 score.
- **Naive Bayes:** Maintains a balanced performance across precision, recall, and F1 (all 0.97), but inference time was not recorded.

Table 1 presents the performance of various phishing detection models. Our model outperforms others in both accuracy and inference time.

Table 1: Comparison on TP-OP benchmark datasets.

Detector	ACC	F1	Precision	Recall	Time (s)
Phishpedia	85.15	52.76	98.84	41.30	0.30
KPD	92.05	91.44	99.92	85.99	1.22
GEPAgent	92.95	92.01	98.59	89.80	12.35
ChatPhish	95.80	95.01	98.10	93.60	0.93
PhishAgent	96.10	96.13	95.24	97.05	2.25
Our Model	**98.13**	**98.00**	**98.50**	**98.90**	**0.0135**

Among all models, Phisher (Our Model) outperforms others with the best accuracy of 98.13%, a F1-score of 98.00%, and a significantly low inference time of 0.0135 seconds. This indicates a great balance between real-time efficiency and detection performance.

While other models, such as PhishAgent[9] and ChatPhish[15], also have good accuracy and recall rates, they lag behind slightly in end-to-end accuracy and run a significantly longer inference time for each test case. GEPAgent[11], although providing fair accuracy rates, is very computational with an over 12-second inference time. It has high accuracy but very low recall, i.e., it misses a large number of phishing examples. On the other hand, KPD[4] and PhishAgent[9] have a relatively better-balanced tradeoff but cannot be compared with the overall performance of our model. Overall, Phisher excels over current detectors in virtually all aspects, becoming a reliable, stable, and efficient solution for real-time phishing detection.

4 Discussion

In this paper, we introduced an end-to-end multimodal phishing detection model that seamlessly combines visual, textual, and URL-level signals to fight against advanced phishing attacks. Integrating ResNet50-extracted image embeddings, BERT-based HTML content embeddings, and engineered URL features, our system derives an overall multimodal representation of phishing sites in diverse views. Comprehensive experiment on a range of model architectures, such as neural networks, Random Forest, and XGBoost, verified that multimodal fusion enhances classification performance over unimodal baselines. Our top-performing model achieved outstanding accuracy, precision, recall, and F1-score, and a correspondingly competitive inference time, ready for deployment in real time. Our finding justifies the usefulness of multimodal learning in computer security, particular phishing detection, in which malicious users often trick neither just the content nor the link, but the appearance of websites.

The model holds great promise for use in browser extensions, mail gateways, or organizational-level defense systems against phishing. One direction for future work involves transformer-based fusion architectures, dynamic tracking of URLs, and large-scale in-the-wild deployment. Another significant direction involves the extension of multimodal phishing detection to emails, where attackers often use malicious sender addresses, malignant attachments, embedded images, and social engineering in the text of the message. Adding modalities like the analysis of the email header, attachment scanning, and NLP-based detection of language that is persuasive would greatly expand the system's utility. Another promising direction involves cross-channel detection, in which the system correlates signals across websites, emails, and messaging services to capture phishing campaigns at the level of the ecosystem.

Moreover, the use of graph-based modeling to link domains, IP addresses, and email origins could strengthen detection against coordinated attacks. Future efforts could also involve federated learning approaches to enable organization-wide defenses without compromising user privacy, and continuous adaptation through online learning to keep pace with evolving phishing strategies. Finally, deploying the model in real-world enterprise and consumer environments will be critical to validate scalability, user experience, and robustness against adversarial evasion techniques.

Our solution, in general, offers a promising direction toward making the internet safer through smart, multifaceted AI systems that provide protection across websites as well as emails, thereby offering an all-round defense against phishing in today's digital landscape.

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